



SunSense RS Solar Monitoring System

ENG207: Programming

College of Engineering & Energy

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Introduction

Have you ever considered what would happen if the electricity suddenly went out in Kuwait without any warning? This is where the idea for our project began. We saw in recent events that electricity was under threat, and we heard that most developed countries did not seem concerned about what might happen to their power grids. We were surprised and began searching for the source of this confidence. The answer is the use of solar panels. In modern engineering systems, solar panels are commonly used in engineering sectors. Solar panels are implemented to convert sunlight into electrical energy especially in harsh weather regions like Kuwait where the radiation is high. However, the solar panels can be affected by environmental conditions such as temperature. SunSense assists in efficiently calculating Kuwait's natural temperature and humidity levels, which can affect efficiency. It can also address situations where excessive heat causes the system to shut down. This system aims to help Kuwait in the future to develop a solar panel system capable of adapting to all of Kuwait's weather conditions.

Program Design

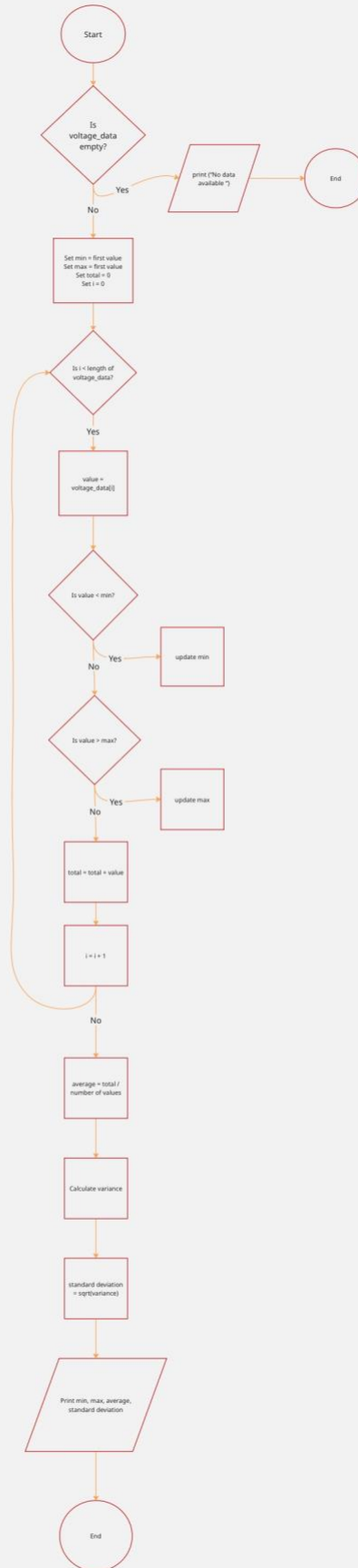
Explanation of the program logic:

The program is designed to simulate and analyse a solar panel system using a modular approach. Its organization into several user defined functions, each of which is in charge of a certain task, enhances the structure of the code. The user can choose between several tasks (1-8) including data generation, reading from a file, analysis, comparing scenarios and visualization. SunSense RS supports both random data generation and reading data from a CSV file, giving users flexibility. Overall, the program follows a clear logical flow, where data is processed through different stages including viewing, statistical analysis, monitoring and floating insuring an organized and efficient system design.

A. Flowchart(s)

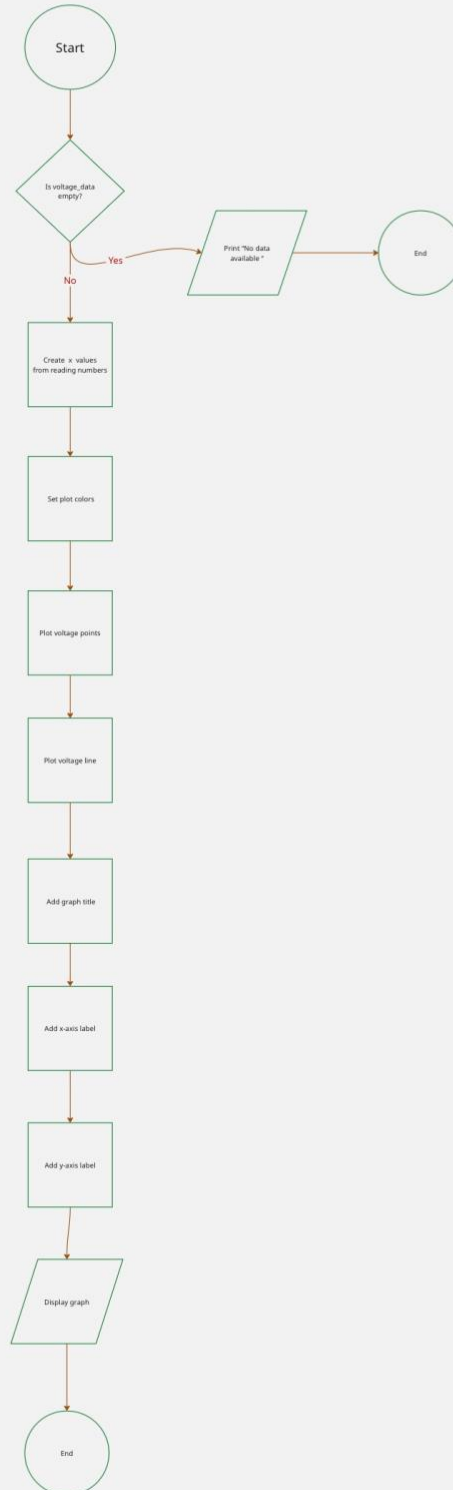


def calculate_statistics



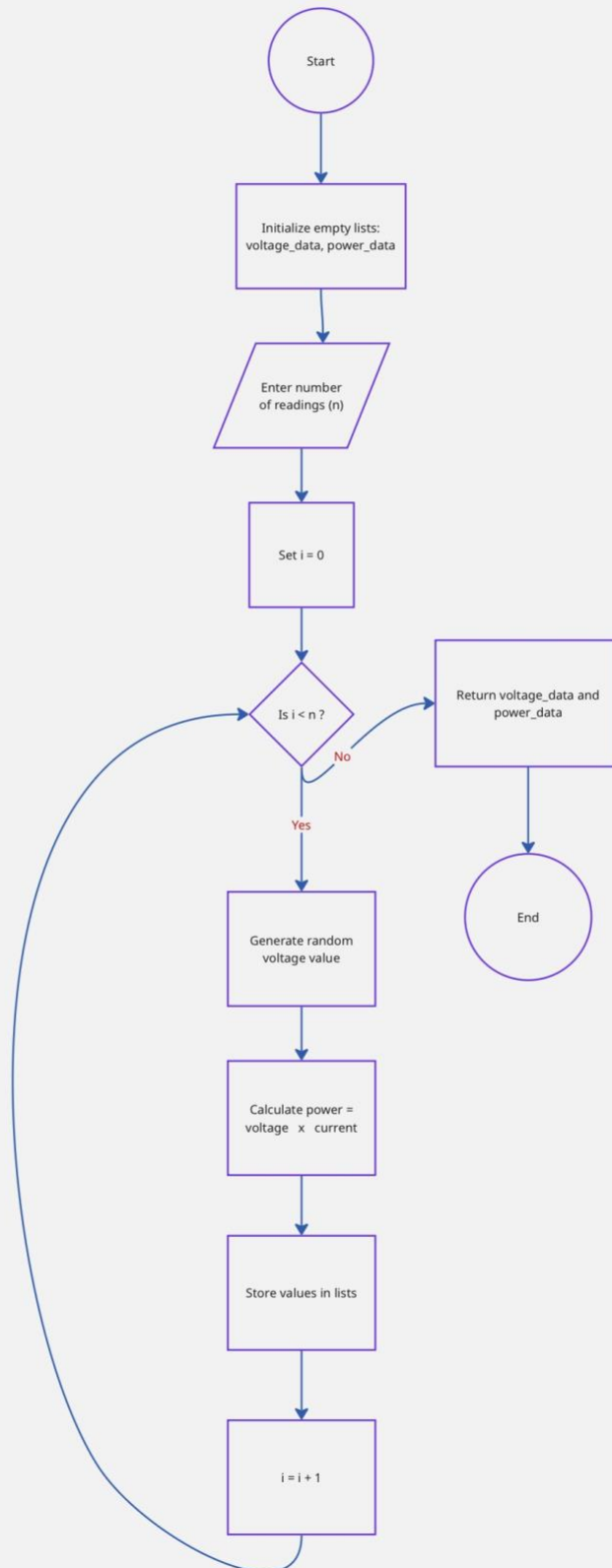


def Plot_data



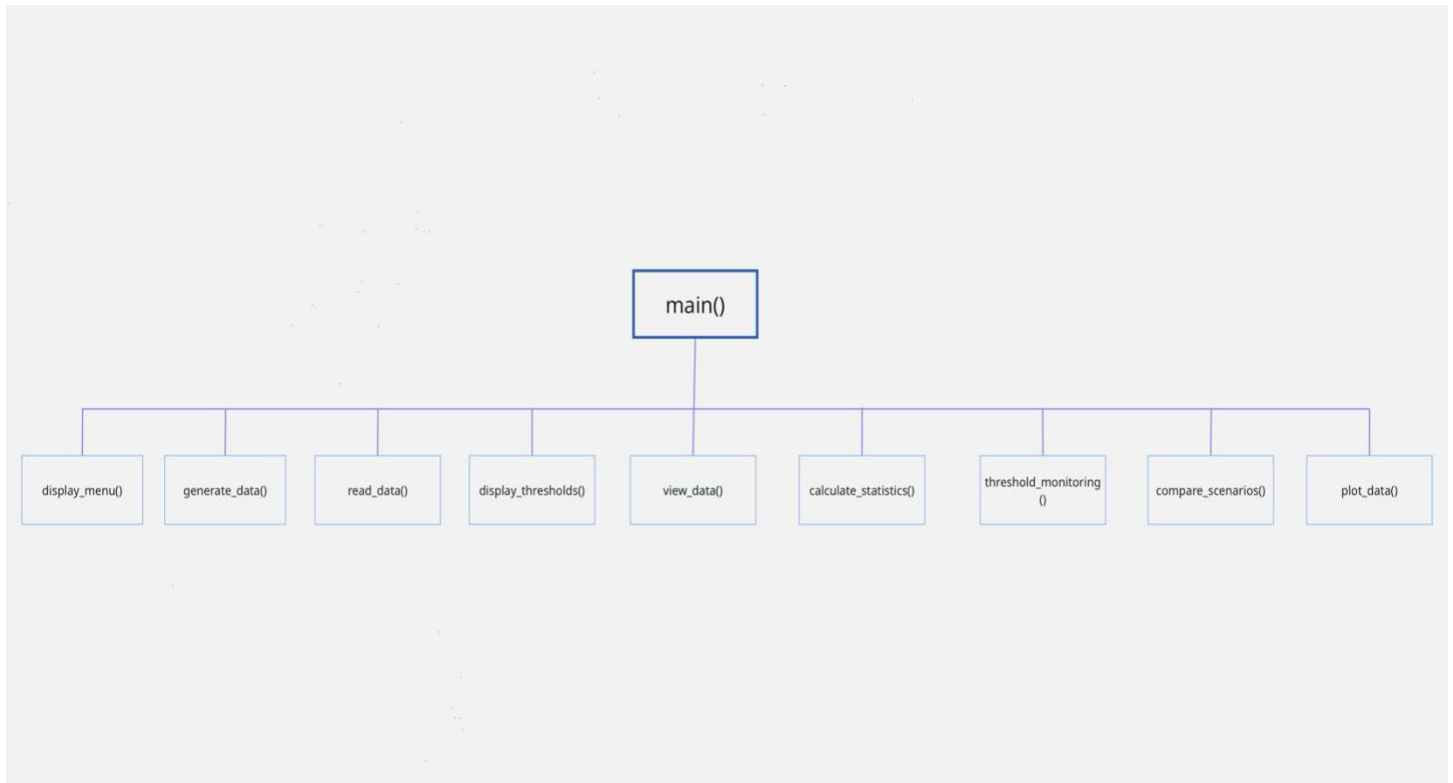


def Generate_data





Hierarchy Chart



The hierarchy chart illustrates the overall structure of the SunSense RS System. The main() function controls the program execution and calls all the other functions, including data generation, data loading, threshold display, data viewing, statistical analysis, threshold monitoring, scenario comparison, and data plotting.

B. User Defined Functions

The program is built using a collection of user-defined functions using Python language, each responsible for performing a specific task. This approach makes the program easier to maintain and enhance code organization.

Firstly, the display menu function is responsible for presenting the main interface of the SunSense RS System. It displays a structured list of options from (1-8) that guide the user through different functions such as data generation, visualization , analysis, and monitoring. The menu is repeatedly shown within a loop until user chooses to exit. Secondly, the generate data function simulates solar sensor readings by generating random voltage values between 10V and 22V. Then it's calculated the Power using the formula $P = V * I$, assuming that the current of SunSense System is fixed and $I = 10A$.

Furthermore, the read data function is for reading voltage values from a CSV file. It processes each value from the file and converts it into a numerical form. Also, it's calculated the power using the same formula. The results are stored in lists to be used by other functions in the program. In the other hand, the view data function is responsible for displaying the collected data in a clear and organized format. It takes the values of the voltage and power lists as input and checks if data available. If the data are not available, it prompts the user to generate or load data. Otherwise, it displays each reading along with the corresponding power value in a readable form.

In addition, the calculate statistics function analyses the voltage data by calculating key statistical values such as minimum, maximum, average and the standard deviation. It checks if data exists at the first then it's computing the required values and displays them in a structured format. Meanwhile, the threshold monitoring function evaluates voltage readings and classifies them into Normal, Warning, and critical readings based on the threshold ranges. Also, it counts the number of warning and critical readings to provide better insight into system performance.

Moreover, the compare scenario's function simulates the behaviour of the solar panel system under different environmental factors in Kuwait, such as normal weather, humid conditions in Kuwait and fault conditions caused by extreme temperature. It generates sample data for each and compares their average values and performance. In addition, the display threshold's function presents the predefined voltage ranges used to classify system conditions. Therefore, it helps users better understand how readings are ranked during the monitoring process.

Finally, the plot data function is used to visualize voltage and power data using graphs. It helps in understanding trends and the overall system behaviour . At last, the main function manages the overall program flow and user interaction. It displays the menu, processes user input and calls the appropriate functions based on selected option and the program continues running until the user chooses to exit.

C. Libraries

Several Python libraries were used to perform system tasks professionally and efficiently. To begin with, the system used the random library to generate random voltage values, which simulate real sensor data for testing purposes. Moreover, the math library was used, which was essential for performing calculations that would be difficult to do manually, thus saving time and speeding up the process. Square roots were specifically used, which were needed to calculate the standard deviation. In addition, matplotlib.pyplot was used, which was crucial in creating the voltage reading graph, allowing the system to display the voltage reading graphically for better analysis and understanding. Lastly, the CSV library was used to read data from CSV files, enabling the system to load real or previously recorded sensor data.

Sample Output and Results Description

Figures A : Sample output of SunSense RS System using Random data

```

=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 1
Choose data source
1.Random Data
2. csv File
Enter your choice (1-2): 1
Enter number of reading 10
Data generated successfully!
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 2
----- Threshold Levels -----
Normal: 15V - 22V
Warning: 18V - 15V
Critical: Below 18V or Above 22V
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 3
Reading 1 : Voltage = 16V, Power = 160W
Reading 2 : Voltage = 17V, Power = 170W
Reading 3 : Voltage = 18V, Power = 180W
Reading 4 : Voltage = 19V, Power = 190W
Reading 5 : Voltage = 20V, Power = 200W
Reading 6 : Voltage = 21V, Power = 210W
Reading 7 : Voltage = 22V, Power = 220W
Reading 8 : Voltage = 23V, Power = 230W
Reading 9 : Voltage = 24V, Power = 240W
Reading 10 : Voltage = 25V, Power = 250W
Data viewed , successfully!
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data

```

(1)



```

measuring v : voltage = 11v, power = 11w
Data viewed , successfully!

Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 4
Statistics:
Minimum Voltage: 18V
Maximum Voltage: 21V
Average Voltage: 14.89
Standard Deviation: 3.48
Statistics calculated successfully!

Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 5
Number of Warning Readings: 4
Number of Critical Readings: 0
Threshold Monitoring completed

Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 6
--- Scenario Comparison ---
Normal Average : 18.8
Humidity Average : 12.6
Fault Average : 14.8

Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 

```

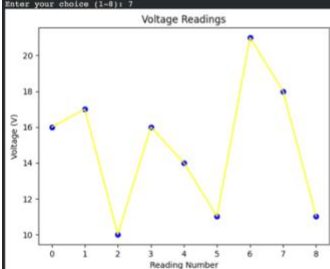
(2)

```

3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 6
--- Scenario Comparison ---
Normal Average : 18.8
Humidity Average : 12.6
Fault Average : 14.8

Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 7

```



```

=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 8
Exiting the program , Goodbye!

```

(3)

The figures above shows that execution of the program starting with generating random voltage data . The program applies different functions starting with generate data function , where random voltage values create and power is also calculated. Then display thresholds function is used to display the predefined voltage ranges. After that , view data function displays the readings. The calculate statistics function calculates the minimum, maximum, average and standard deviation. Next, thresholds monitoring function classifies the reading into Normal, Warning and Critical. Then the scenarios was compared using compare scenarios function and plot data function visualize the results in graphs. Finally the user exits the SunSense system .

Figures B : Sample output SunSense RS using Data from CSV File

```

=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 3
Reading 1 : Voltage = 18.0V, Power = 100.0W
Reading 2 : Voltage = 20.0V, Power = 200.0W
Reading 3 : Voltage = 14.0V, Power = 140.0W
Reading 4 : Voltage = 22.0V, Power = 220.0W
Reading 5 : Voltage = 10.0V, Power = 100.0W
Data viewed , successfully!
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 4
Statistics:
Minimum Voltage: 10.0V
Maximum Voltage: 22.0V
Average Voltage: 16.80
Standard Deviation: 4.31
Statistics calculated successfully!
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 5
Number of Warning Readings: 2
Number of Critical Readings: 0
Threshold Monitoring completed
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8):

```



```
Average voltage: 10.4v
Standard Deviation: 4.31
Statistics calculated successfully!
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 5
Number of Warning Readings: 2
Number of Critical Readings: 0
Threshold Monitoring completed
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 2
----- Threshold Levels -----
Normal: 15V - 22V
Warning: 10V - 15V
Critical: Below 10V or Above 22V
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 6
----- Scenario Comparison -----
Normal Average : 19.8
Humidity Average : 13.4
Fault Average : 13.6
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8):
```

(5)

```
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 6
----- Scenario Comparison -----
Normal Average : 19.8
Humidity Average : 12.6
Fault Average : 14.0
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 7
```

```
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 8
Exiting the program , Goodbye!
```

(6)



Figures B shows the execution of the system using both generate data and read data functions. In the first case we used generate data function is used to create random values for voltage and read data function is used to reads voltage values form CSV file and computes the power. Then the program performs the same as in random values. Furthermore, the program was implemented in both Jupyter Notebook and Google Colab . In Jupyter we used read data function to read from CSV file . On the other hand, in Google Colab generate data function used to provide two options for the user either generate random values or reading from file using if-elif statements.

Figures C : SunSense RS System Sample Output for Invalid input

```

=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 10
Invalid choice. Please try again.
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 

```

(7)

This figure shows that when the user enters an invalid number option the system displays an error message. However, the program continues until user select number 8 to exit.



```
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 1
Choose data source
6.Random Data
1. csv File
Enter your choice (G-L): k
Invalid choice!
=====
Welcom To SunSense RS Solar Panle System,please select an option:
=====
1.Generate Solar sensors Data
2.Set Threshold Levels
3.View Raw Data
4.Calculate Stat
5. Threshold Monitoring
6.Compare Scenarios
7.Plot Results
8.Exit
Enter your choice (1-8): 8
Exiting the program , Goodbye!
```

(8)

This figure shows that if the user enters an invalid character while selecting the data source. Also, the system prints an error message to notify the user that the selected character was wrong. The system allows the user to try again.

Conclusion

In conclusion, the idea for this system was inspired by recent events in Kuwait, where the country may face power outages if generators fail. The project focuses on developing a system capable of adapting to Kuwait's fluctuating weather, enabling solar panels to operate efficiently. The system, named SunSense RS, was designed to monitor and analyse solar panel performance. SunSense represents a system for utilizing solar energy effectively, while RS refers to the initials of the developers. In summary, the goal of this project is to provide users with multiple options to calculate voltage and power, and to determine whether the output is within the normal operating range of solar panels under different environmental conditions, such as normal temperatures, high humidity, and extreme heat, which are common in Kuwait's climate.

The lessons learned from this project were numerous. The most important lesson was patience, as a failed attempt can become motivation to achieve the goal. This project required significant patience, especially during the early stages of the coding process, where many challenges were encountered. This was particularly difficult because the project focused on Kuwait, making it difficult to obtain accurate information regarding the value of the volt or anything else. However, we learned that success begins with an idea, and that idea can develop into a meaningful project that benefits society. While challenges are inevitable, using them as opportunities for improvement is essential. This experience also highlighted the importance of planning and choosing the right partner, as both have a significant impact on the efficiency and success of the work.

We faced several challenges, including code errors that we could not identify. However, using available resources helped us resolve these issues. Over time, we gained experience. We also had to learn to work on the code using our own computers. Additionally, we encountered a problem when reading the file in Jupyter, as it required a specific function to properly handle the data. We used another computer and Google Colab, but we also faced the challenge of running the file's cell before the user could select their preferred option, such as reading from the file or generating random data. Using different platforms proved to be the most suitable approach for each of us. Finally, for future improvements, the system could be enhanced by integrating real time sensor data instead of simulated inputs and improving the user interface. This would make the system more accurate to real world solar energy applications.

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